Delta Dependency Prioritizes Paralog-Based Synthetic Lethality Candidates Across Solid Tumor Types

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Abstract

Background: Paralog compensation is a specific mechanism of synthetic lethality (SL): when a tumor-suppressor driver gene is lost to mutation, its sequence-similar paralog can become conditionally essential. Systematic prioritization of these paralog-SL candidates remains limited because existing computational methods function as black boxes that cannot explain their predictions, making it difficult to design targeted validation experiments.

Results: We introduce Delta Dependency (DD), a single interpretable metric that measures the dependency shift on a paralog gene between driver-mutant and wild-type cell lines in DepMap (1,208 lines, 23 solid tumor types). In the pre-specified primary evaluation — a lineage-level benchmark on twelve curated, evidence-tiered paralog-SL pairs — DD achieved AUROC = 0.676 (0.725 after excluding two pairs with DepMap-derived labels; 1.000 on the Tier A ∪ Tier B external benchmark, on which two of five pairs were lineage-evaluable). In a pair-level benchmark on the same 72 test pairs, the interpretable composite score reached AUROC = 0.831, statistically indistinguishable from the best of four standard multi-feature classifiers (SVM-RBF, 0.843) under leave-one-pair-out cross-validation. Exploratory analyses across seven CPTAC cohorts ($n = 771$) detected protein-level paralog co-variation invisible at the RNA level. Dependency-window scoring prioritized ARID1A→ARID1B (DWS = 2.82) as the leading selective candidate, suggesting a hypothesis-generating biomarker strategy centered on truncating ARID1A mutations.

Conclusions: DD offers a mechanistically transparent alternative to black-box SL prediction. Each DD nomination traces to a measured dependency shift, enabling rational experimental follow-up. DD is an association-based prioritization metric rather than a causal test of synthetic lethality, and all candidates require experimental validation. An open-source R package (`paralogSL`) and reproducible pipeline are provided.

Keywords: Synthetic lethality, paralog compensation, DepMap, CPTAC proteomics, Delta Dependency, dependency window

Background

The PARP inhibitor story crystallizes what makes synthetic lethality (SL) so compelling: BRCA1/2-mutant tumors depend on PARP-mediated repair, and drugging PARP kills them while sparing normal cells [1–5]. The logic is elegant; repeating it for other mutations has proven hard. Fewer than 30,000 human SL interactions are catalogued in SynLethDB [6], and the existing computational prediction methods — graph neural networks [7, 8], graph convolutional networks with dual dropout [9], graph-regularized matrix factorization [10], knowledge-graph neural networks [11], and negative-sample-free contrastive learning [12] — train almost entirely on these limited known pairs. Feng et al. [13] showed that these methods degrade sharply from the CV1 setting (AUROC > 0.90 for the best methods) to the CV3 gene-pair isolation setting, where most fall to near-random performance (AUROC ≈ 0.5) and the best published result reaches 0.790 [14] — precisely the setting where predictive models matter: nominating genuinely new SL pairs.

Paralog compensation provides a simpler route into this problem [15, 16]. Large-scale analyses of CRISPR dependency data have since established paralog pairs as the most enriched class of synthetic-lethal interactions and shown that interpretable models built on protein-interaction and evolutionary-conservation features can predict them [43]; combinatorial knockout screens have begun to map paralog synthetic lethality directly [44, 46]. Gene duplication events leave behind paralogs — sequence-similar gene pairs with overlapping biochemical roles. When a paralog acting as a tumor suppressor (the "driver") is inactivated by mutation, its partner paralog can become conditionally essential. This is *sequence-paralog SL*: synthetic lethality between two genes that share high protein sequence identity (operationalized with a validated *k*-mer identity-enrichment proxy; Methods) and conserved functional domains. Known examples include *SMARCA4*→*SMARCA2* (SWI/SNF ATPase subunits) [17], *ARID1A*→*ARID1B* (BAF complex subunits) [18, 19], *EP300*→*CREBBP* (histone acetyltransferases) [20], *PIK3CA*→*PIK3CB* (PI3K catalytic isoforms) [6], *AKT1*→*AKT2*, *CCNE1*→*CCNE2*, *CDK4*→*CDK6*, and *MAP2K1*→*MAP2K2* [6]. The strength of evidence behind these catalogued pairs varies — from direct genotype-conditional experiments to redundancy and pharmacologic data — so we verified each primary citation and assigned every pair an evidence tier (Supplementary Table S3; Results). In every case, the driver and paralog share sequence homology and biochemical function; when the driver is inactivated by mutation, the remaining paralog becomes the cancer cell's dependency.

Two related phenomena need to be separated from sequence-paralog SL. *Functional analogs* such as BRCA1 and BRCA2 share pathway roles (homologous recombination repair) but have no sequence homology. *Pathway-level SL* interactions, such as BRCA1↔PARP1 and PTEN↔PIK3CB [21, 22], exploit indirect pathway wiring rather than direct biochemical redundancy. Our analysis concentrates on sequence-paralog pairs; functional analogs including BRCA1/BRCA2 are retained as mechanistic comparators (specificity references, not verified negatives) and flagged throughout (Supplementary Table S3).

D'Antonio et al. published the first systematic analysis of negative genetic interactions between functional paralogs in cancer cell lines, identifying buffering relationships at the gene-family level [16]. That study established the landscape but did not benchmark paralog-based predictions against published SL methods, did not incorporate proteomic validation, and did not address clinical stratification or therapeutic window assessment. Three practical gaps remain. Current methods are black boxes: when a neural network predicts SL, there is no way to trace *why*, which blocks rational experimental design. The optimal readout for paralog compensation is unsettled, since RNA-level and protein-level signals can diverge, and there has been no systematic proteomic survey. Finally, no study has layered orthogonal evidence (proteomics, drug sensitivity, clinical biomarkers, structure) onto genomic prediction to move from a ranked list to a prioritized, experimentally actionable nomination.

Here we introduce Delta Dependency (DD): the difference in mean Chronos gene-effect scores between driver-mutant and wild-type cell lines. DD replaces black-box classification with a single subtraction: $|\text{DD}| = 0.386$ for ARID1A→ARID1B in ovarian cancer cells means the paralog's essentiality score shifts by 0.386 units when the driver is mutant, a measurable quantity that can guide experimental design. We evaluate DD across 23 solid tumor types using a 40-gene driver panel (Supplementary Table S5) curated from TCGA PanCanAtlas driver analyses [23] and the COSMIC Cancer Gene Census [42], and we support the genomic predictions with CPTAC proteomics (seven cohorts), PRISM drug screening (1,482 compounds), MSI-aware patient stratification, mutation-type analysis, therapeutic window quantification, and structural druggability assessment.

Results

DD defines a dependency-shift metric for paralog-SL prioritization

We defined Delta Dependency as:

$$\mathrm{DD}(D,P,c) = \bar{G}_{P,D\text{-WT},c} - \bar{G}_{P,D\text{-MUT},c} \tag{1}$$

where $\bar{G}$ denotes the mean Chronos gene-effect score, D is the driver gene, P the candidate paralog, and c the cancer lineage. Because lower Chronos gene-effect scores indicate stronger dependency, positive DD indicates increased dependency on paralog P in driver-mutant cell lines relative to wild-type within lineage c, consistent with paralog compensation.

Across 66,595 HGNC paralog pairs and 1,208 DepMap cell lines, DD exceeded AUROC = 0.7 in 7 of 8 evaluable lineages on the primary frame (Fig. 1a; Supplementary Table S1): Esophagogastric (AUROC = 0.965, $n = 50$ pairs), small cell lung cancer (SCLC; 0.906), Bladder Urothelial (0.844), Colorectal (0.828), Endometrial (0.818), Breast (0.750), and non-small cell lung cancer (NSCLC; 0.741); Ovarian (0.661) fell below this threshold. Under the relaxed ≥3-per-group sensitivity frame (12 evaluable lineages), 9 of 12 exceeded 0.7, with Biliary Tract (0.990) and Pancreatic (0.949) among the strongest performers. Performance did not differ significantly by oncogenic mechanism (Fig. 1b; sensitivity frame, because the primary frame retains only one oncogene-driven lineage): TSG-driven cancers had a mean AUROC of 0.814 ($n = 9$ lineages) versus 0.768 for oncogene-driven cancers ($n = 3$ lineages; permutation $p = 0.645$, exact Mann-Whitney $p = 0.600$). The direction of the difference — numerically stronger compensation signals in TSG-driven cancers, where paralog compensation arises from loss-of-function mutations — is biologically expected, but with only three oncogene-driven lineages the comparison is severely underpowered and we report it as descriptive.

We compared DD against eight published SL prediction methods in the CV3 setting reported by Feng et al. (2024) [13] (Fig. 1c; Table 1). DD achieved AUROC = 0.676 without any training, below the best published result in that setting (SLMGAE [14], 0.790), though we stress that the evaluation frameworks are analogous, not identical. We therefore treat the published values as contextual reference points rather than a formal benchmark. For a true head-to-head comparison on the same 72 paralog pairs (6 known positives), we trained four standard classifiers — logistic regression (LR), random forest (RF), SVM-RBF, and SVM-Linear — on five features (|DD|, PCS, ΔExpression, necessity, mutation frequency; the composite score was excluded from classifier inputs because it is a deterministic function of the other features) under leave-one-pair-out cross-validation. The composite score (AUROC = 0.831) and SVM-RBF (0.843) performed best and are statistically indistinguishable at this sample size, followed by random forest (0.722) and DD alone (0.566); the linear classifiers were unstable (SVM-Linear 0.114, LR 0.136). With only 6 positives, multi-feature training amplifies noise and no classifier result should be over-interpreted; in the full LR fit no individual feature reached significance (all five feature

p-values > 0.23). On the same 72-pair frame, the composite score reached AUPRC $= 0.356$ ($4.3\times$ the baseline prevalence of 0.083), confirming enrichment beyond random expectation. These comparisons are confined to the paralog-based SL task; DD is not intended for pathway-level interactions such as BRCA1↔PARP1. A DD + k-mer identity-enrichment filter ($\geq$30% proxy) achieved AUROC $= 1.000$ on a high-identity subset of 3 pairs (2 known SL), but with only two positives this is anecdotal and we report it without a significance claim. Unlike neural networks, every DD nomination maps to a single measured quantity — a dependency shift — which makes it straightforward to design a validation experiment.

Component decomposition on the same test set: DD (AUROC $= 0.676$) versus PCS (0.825), ΔExpression alone (0.547), and necessity (0.642; Fig. 1d). A paired bootstrap over entries found no significant difference between DD and any other component (PCS minus DD $+0.150$, 95% CI -0.110 to $+0.456$). PCS in particular carries substantial signal, which motivates the composite score.

We evaluated DD on evidence-tiered test sets (Table S3), reclassifying the twelve curated pairs after verifying each primary citation. Three pairs have direct genetic synthetic-lethal evidence from dual-gene perturbation assays (*Tier A*: AKT1→AKT2 [38], CDK4→CDK6 and MAP2K1→MAP2K2 [46]). Two further pairs are demonstrated selective dependencies in driver-mutant cells with functional validation (*Tier B*: SMARCA4→SMARCA2 [17], ARID1A→ARID1B [18]); the Tier A ∪ Tier B set constitutes the pre-specified primary external benchmark. Five pairs rest on indirect, reciprocal-direction-only, or DepMap-derived evidence (*Tier C*: EP300→CREBBP [20, 41], PIK3CA→PIK3CB [22], CCNE1→CCNE2 [39], FBXW7→FBXW2, PPP2R1A→PPP2R1B), and two serve as mechanistic comparators (BRCA1↔BRCA2, STK11→SIK1). On the full curated set of twelve pairs DD achieved AUROC $= 0.676$ (110 driver×paralog×lineage entries, 8 positives), and this estimate proved robust to label-quality concerns: excluding the two DepMap-derived pairs raised AUROC to 0.725, restricting to the eight pairs with pre-DepMap experimental evidence gave 0.774, and a direction-strict analysis relabelling the EP300→CREBBP entries as non-positive left it unchanged at 0.676. On the primary Tier A ∪ Tier B frame, both evaluable positive entries (ARID1A→ARID1B in Endometrial and Ovarian cancers) ranked above all 102 unlabeled control entries (AUROC $= 1.000$); SMARCA4→SMARCA2 and the three Tier A pairs had too few qualifying driver-mutant cell lines for lineage-level evaluation under the $\geq$5-per-group rule. With at most 2–3 strictly validated positives, AUROC computed on small subsets is not informative, so we rely on the full-set estimate and supplement it with effect-size ranking: ARID1A→ARID1B has the largest lineage-level DD (0.386 in Ovarian cancer; Hedges' $g = 1.39$; Supplementary Table S8) and a top-tier dependency-window and selectivity profile (Table 2), providing orthogonal prioritization beyond classification performance.

Leave-one-lineage-out AUROC was stable (range 0.656–0.704). Bootstrap resampling (1,000 iterations, 72 pairs, 6 positives) gave a 95% CI of 0.185–0.813 (Supplementary Fig. S8). In the pair-level aggregated framework, the null distribution from 10,000 label permutations was centered at 0.501 and the observed AUROC (0.500) did not exceed it (empirical $p = 0.503$; Supplementary Fig. S8): with only 6 positives the aggregated per-pair evaluation carries no detectable signal, so we treat the lineage-level evaluation as the primary framework throughout. We estimate $\geq$25 validated pairs would be needed for 80% power at $\alpha = 0.05$. Copy number alterations explained $<$10% of gene-effect variance for all 23 paralog genes tested (maximum $R^2 = 0.057$; Supplementary Fig. S4), and multivariate regression controlling for CNV left DD estimates virtually unchanged (mean $|\Delta DD| = 0.005$). ARID1A→ARID1B remained highly significant after CNV adjustment (adjusted $p = 6.5 \times 10^{-42}$), as did SMARCA4→SMARCA2 ($p = 3.5 \times 10^{-29}$) and EP300→CREBBP ($p = 1.4 \times 10^{-13}$). Controlling for expression ($G_P \sim D_{\text{mutation}} + \text{Expr}_P$) produced similar results (mean $|\Delta\text{DD}| = 0.014$); significance was retained by the strongest pairs (ARID1A→ARID1B $p = 4.1 \times 10^{-41}$, PIK3CA→PIK3CB $p = 1.5 \times 10^{-11}$, EP300→CREBBP $p = 6.3 \times 10^{-13}$), whereas weaker pairs did not reach significance in the pan-cancer frame, consistent with lineage-specific compensation. TP53 co-mutation, present in 76/121 ARID1A-mutant lines, had negligible impact on ARID1A→ARID1B ($\Delta\text{DD} = 0.0004$, adjusted $p = 5.8 \times 10^{-42}$). Adding lineage fixed effects attenuated but preserved the association (lineage-adjusted $p = 2.2 \times 10^{-13}$; TP53- and CNV-adjusted with lineage fixed effects $p = 7.0 \times 10^{-13}$).

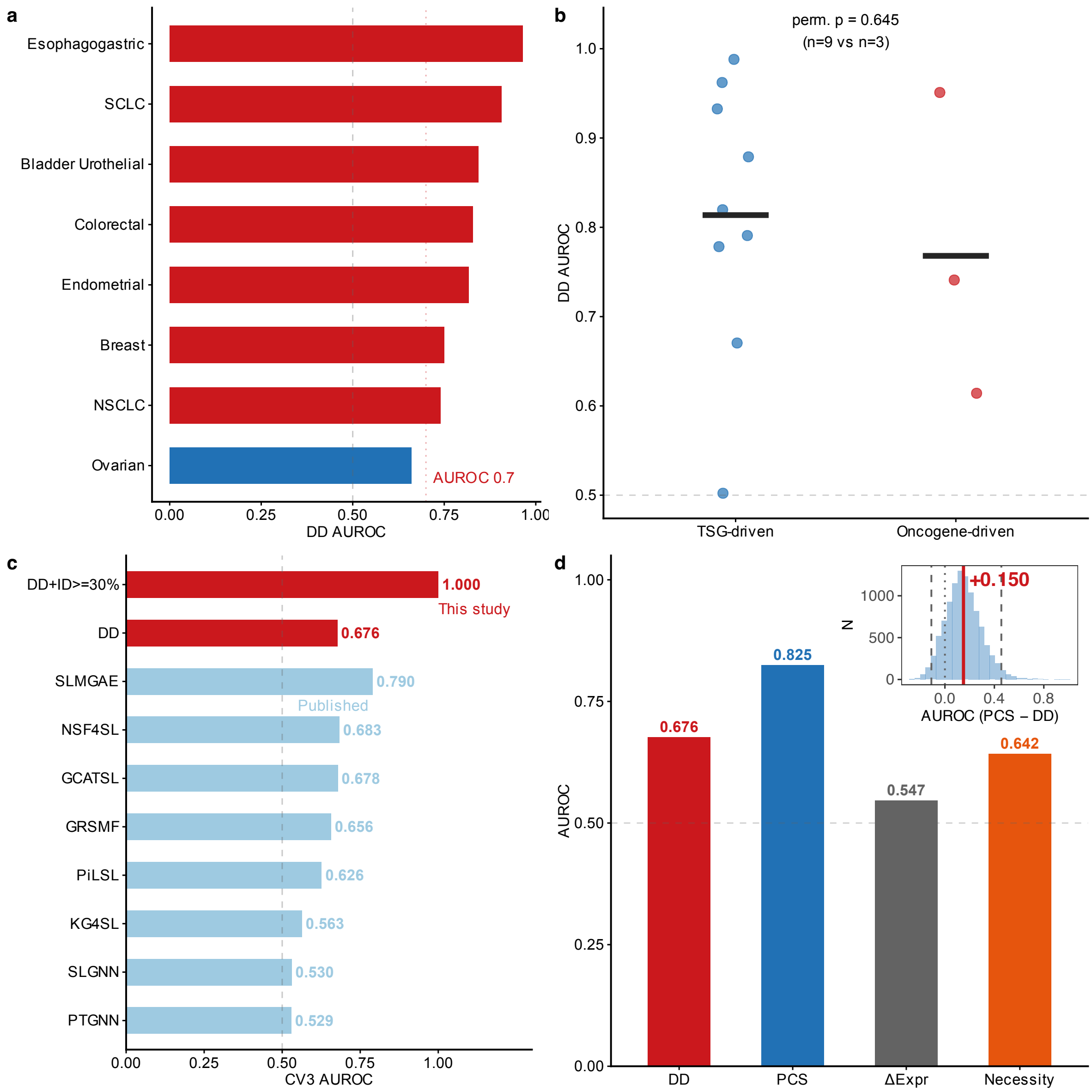

Fig. 1: Delta Dependency: pan-cancer performance, benchmark, and validation. **a**, DD AUROC across 8 evaluable solid tumor types on the primary $\geq$5-per-group frame (red: AUROC $>$ 0.7; blue: 0.5–0.7). Dashed line, random classifier (0.5). Bar labels show n tested pairs. Sample sizes and full lineage names in Supplementary Table S1; the $\geq$3-per-group sensitivity frame yields 12 evaluable lineages. **b**, DD performance stratified by oncogenic mechanism. TSG-driven tumors ($n = 9$) show numerically higher AUROC than oncogene-driven tumors ($n = 3$; permutation $p = 0.645$, not significant). Points, individual lineages; center line, group mean. Computed on the $\geq$3-per-group sensitivity frame because the primary $\geq$5 frame retains only one oncogene-driven lineage. **c**, Contextual comparison: DD AUROC vs. published CV3 values from eight deep learning methods [13]. Red bars, this study; blue bars, published values. *Note: different test sets; not a head-to-head benchmark.* The interpretable composite score (AUROC = 0.831) matches the best of four multi-feature classifiers tested under leave-one-pair-out cross-validation (SVM-RBF, 0.843; classifier range 0.114–0.843), and each DD-based nomination traces to a single measurable quantity (|DD|), enabling direct experimental design, whereas multi-feature classifiers produce opaque scores. **d**, Component decomposition: DD (AUROC = 0.676) versus PCS (0.825), ΔExpression alone (0.547), and necessity (0.642). Inset: paired bootstrap distribution of PCS minus DD (10,000 resamples; solid line, mean +0.150; dashed lines, 95% CI −0.110 to +0.456; the difference is not significant). The pair-level bootstrap and negative control are in Supplementary Fig. S8.

Table 1: DD performance in context of published CV3 results (not a head-to-head benchmark). Published values are CV3 (gene-pair isolation) AUROCs from Feng et al. (2024), Supplementary Data 1 (NSMRand negative sampling, 1:1 positive:negative ratio, complete dataset). DD and DD + ID values from this study: DD was evaluated on the full lineage-level frame (110 driver–paralog–lineage entries, 8 positives); DD + ID on the high-identity subset (3 pairs, 2 positives) and is reported as anecdotal. *Evaluation frameworks differ: published methods were tested on general SL gene-pair universes; DD was tested on paralog-SL pairs only. Direct AUROC comparison is not warranted.* Interpretability: whether predictions can be directly traced to specific biological features.

Method	CV3 AUROC	Interpretability
DD + ID $\geq$ 30% (this study)	1.000	High
DD (this study)	0.676	High
SLMGAE [14]	0.790	Low
NSF4SL [12]	0.683	Low
GCATSL [8]	0.678	Low
GRSMF [10]	0.656	Low
PiLSL [24]	0.626	Low
KG4SL [11]	0.563	Low
SLGNN [7]	0.530	Low
PTGNN [25]	0.529	Low

Orthogonal proteomic evidence: protein-level paralog co-variation

Whether paralog compensation acts at the transcript or protein level matters. We analyzed quantitative proteomics from seven independent CPTAC cohorts: BRCA ($n = 122$), COAD ($n = 97$), LUAD ($n = 110$), GBM ($n = 99$), PDAC ($n = 140$), UCEC ($n = 95$), and LUSC ($n = 108$), totaling 771 tumor samples with matched protein abundance [26–28].

Protein-level paralog co-variation was detectable across cohorts (Fig. 2a). EP300↔CREBBP showed the most consistent signal: significant in 5 of 7 cohorts after within-pair BH correction (BRCA: $r = 0.263$, $q = 0.005$; COAD: $r = 0.510$, $q < 10^{-3}$; LUAD: $r = 0.380$, $q < 10^{-3}$; GBM: $r = 0.501$, $q < 10^{-3}$; LUSC: $r = 0.372$, $q < 10^{-3}$; mean $r = 0.320$; Fig. 2b). Across all 26 pairs $\times$ 7 cohorts (122 tests), 42 were significant at global BH $q < 0.05$. KRAS↔HRAS had the strongest single-cohort correlation ($r = 0.667$, $p < 10^{-4}$ in LUSC). Protein co-variation alone does not prove compensatory dependency; it could reflect shared transcriptional regulation, tumor purity, or copy number linkage. These results serve as orthogonal mechanistic support, not causal validation.

The RNA-level picture was much weaker. ΔExpression alone achieved AUROC = 0.547 for discriminating known paralog-SL pairs (Fig. 2c), far below the protein-level co-variation signal and below DD (0.676). This gap — protein detectable, RNA marginal — helps explain why transcript-based methods have underperformed. To test directly for compensatory upregulation, we compared ARID1B protein abundance in ARID1A-mutant ($n = 37$) versus wild-type ($n = 58$) endometrial tumors in the UCEC cohort. ARID1B was 55% higher in mutant tumors (Wilcoxon $p = 0.082$; Fig. 2d). The p-value is suggestive but does not meet $p < 0.05$. The effect size, however, is consistent with a model in which loss of one complex subunit stabilizes its paralog.

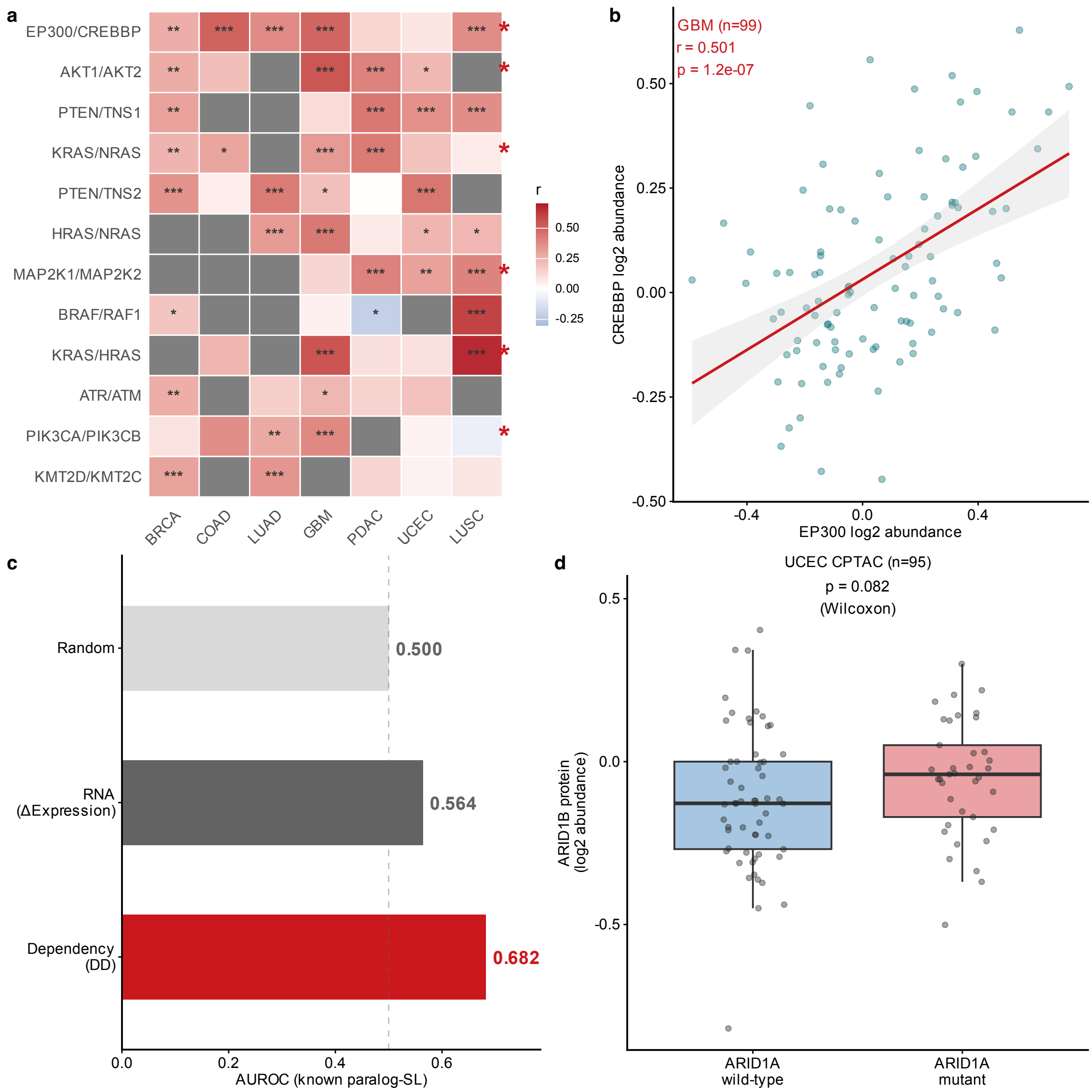

Fig. 2: CPTAC proteomic co-variation across seven cohorts. **a**, Cross-cancer paralog protein co-variation (Pearson's r) across seven independent CPTAC cohorts ($n = 771$ total). Significance: *** $q < 0.001$, ** $q < 0.01$, * $q < 0.05$ (BH-corrected within each pair). Blue cells with significance markers indicate significant *negative* correlations. Known SL pairs marked with $*$. **b**, Representative scatter plot: EP300 vs. CREBBP protein abundance in the GBM cohort ($n = 99$, $r = 0.501$, $p < 10^{-4}$). Gray band, 95% CI of linear regression. **c**, Dependency-based (DD, AUROC = 0.676) vs. RNA-level (ΔExpression, AUROC = 0.547) prediction performance for known paralog-SL pairs. The RNA signal is weak (dashed line, random expectation). **d**, Mutation-conditioned proteomics: ARID1B protein abundance in ARID1A-mutant ($n = 37$) vs. wild-type ($n = 58$) endometrial tumors (UCEC CPTAC cohort). ARID1B was higher in ARID1A-mutant tumors (+55%, Wilcoxon $p = 0.082$), providing suggestive evidence for compensatory protein upregulation upon driver loss.

Clinical context modulates DD signal: MSI status, mutation type, and survival

Microsatellite instability (MSI) is a routine clinical biomarker in colorectal and endometrial cancers, and MSI-H tumors accumulate hypermutation-driven passenger events. We tested whether this mutational background dilutes paralog compensation signals by classifying cell lines with the official DepMap microsatellite annotation (MSIsensor2 MSIscore; MSI-H = score > 20; Methods) and computing DD within each subgroup. Contrary to that expectation, MSI-H subgroups showed numerically stronger paralog-SL predictive signal in both cancer types on the $\geq$3-per-group sensitivity frame (Fig. 3a): endometrial AUROC = 0.838 (MSI-H, $n = 17$) vs. 0.556 (MSS, $n = 11$), $\Delta = +0.282$; colorectal AUROC = 0.767 (MSI-H, $n = 14$) vs. 0.712 (MSS, $n = 45$), $\Delta = +0.055$. On the primary $\geq$5 frame the endometrial subgroups are not evaluable (fewer than two gold-standard pairs) and the colorectal subgroups show no difference (0.574 MSI-H vs. 0.595 MSS), so the MSI contrast is strictly exploratory. Subgroup sizes are modest and only 3–4 gold-standard pairs were evaluable per subgroup, so the differences were not formally tested for significance; these numbers are hypothesis-generating. They nonetheless argue that a hypermutated background does not preclude paralog-SL detection, and that MSI status need not restrict biomarker-enrichment strategies.

Mutation type further modulated the signal. For the leading TSG pair, truncating mutations produced substantially stronger DD than missense variants (ARID1A→ARID1B in Ovarian: DD = 0.388 truncating vs. 0.020 missense, ΔDD = +0.368; Fig. 3b), and truncating-only DD reached AUROC = 0.726 in Breast cancer versus 0.391 for missense-only (Supplementary Fig. S6). The pattern was not universal, however: across all tested pairs, mean |DD| was modestly larger for missense than for truncating subgroups (0.042 vs. 0.026; paired t-test $p = 3 \times 10^{-4}$), driven mainly by oncogene pairs such as PIK3CA→PIK3CB in which hotspot missense mutations define the mutant group (Supplementary Fig. S6b). These are point estimates from small per-subgroup samples and should be treated as descriptive observations.

TCGA PanCan survival analysis of paralog gene expression across breast cancer patients ($n = 1{,}082$) [29] identified significant associations between high paralog expression and shortened overall survival for BRCA2 (HR = 1.116, 95% CI: 1.010–1.234, $p = 0.032$) and ATR (HR = 1.112, 95% CI: 1.006–1.230, $p = 0.039$), with trend-level associations for ARID1B and CRKL (Fig. 3c). These hazard ratios, while statistically significant, represent modest effect sizes (11–12% risk increase) and were not corrected for multiple testing across 8 genes; after Bonferroni correction ($\alpha = 0.05/8 = 0.00625$) neither association would remain significant. Paralog expression alone is unlikely to serve as a strong standalone clinical biomarker but may contribute to multi-gene prognostic signatures. Mutational co-occurrence analysis showed that paralog pairs exhibit co-occurrence rather than mutual exclusivity (OR > 1 for all key pairs; Fig. 3d). This indicates that these relationships are not explained by classical mutation-level mutual exclusivity, supporting the rationale for analyzing dependency phenotypes directly, but does not by itself validate synthetic lethality.

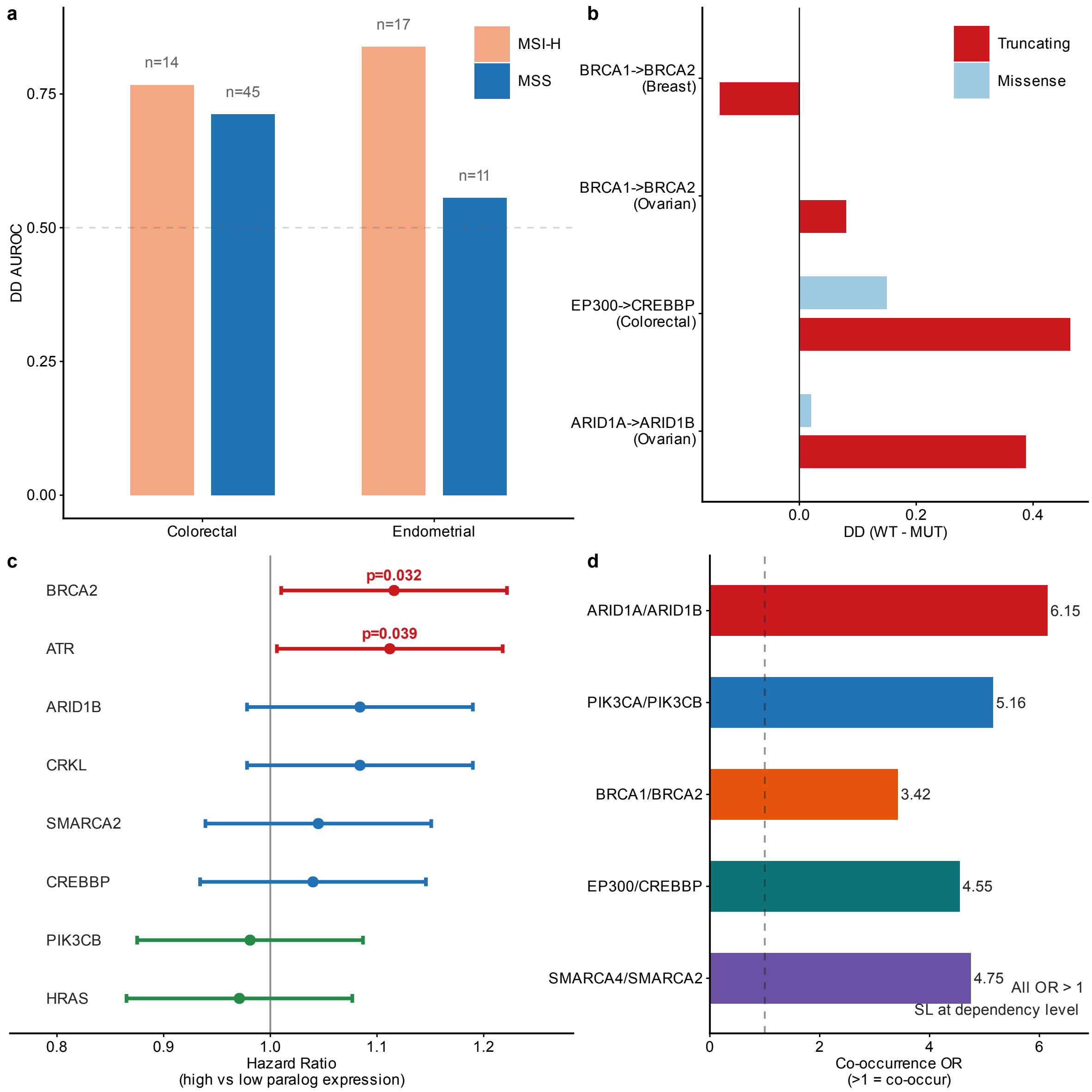

Fig. 3: Clinical stratification by MSI, mutation type, and survival. **a**, DD AUROC for MSI-H vs. MSS subgroups in colorectal and endometrial cancers, using the official DepMap MSIsensor2 annotation (MSIscore > 20 = MSI-H), computed on the $\geq$3-per-group sensitivity frame (the primary $\geq$5 frame is not evaluable for endometrial and shows no colorectal difference; see text). Sample sizes shown above bars. MSI-H subgroups showed numerically stronger DD signal in both cancer types. **b**, DD (WT $-$ MUT) for truncating vs. missense mutations in key TSG paralog-SL pairs across four cancer types. Positive values indicate compensation; larger truncating bars indicate stronger compensation under truncating mutations. See Supplementary Fig. S5 for full results. **c**, Forest plot of paralog gene expression vs. overall survival in TCGA BRCA ($n = 1{,}082$). BRCA2 (HR $= 1.116$, $p = 0.032$) and ATR (HR $= 1.112$, $p = 0.039$) are significant. Error bars, 95% CI. **d**, Mutational co-occurrence odds ratios for key paralog pairs. All pairs show OR > 1 (range: 3.4–6.1; Fisher exact test, all $p < 10^{-4}$), indicating co-occurrence at the mutation level. This supports the need to analyse dependency phenotypes directly but does not by itself establish synthetic lethality.

Therapeutic-window and structural features prioritize ARID1A→ARID1B

To assess pharmacologic relevance, we analyzed the PRISM Repurposing dataset (1,482 compounds across 727 cell lines) [30]. At a discovery-stage threshold (BH $q < 0.25$, ΔAUC< 0, enrichment> 0.1; Supplementary Fig. S5), 633 compound-paralog associations passed, of which roughly 158 are expected false positives given the permissive cutoff. The relaxed threshold is deliberate because this is a screen, not a confirmatory analysis. Encouragingly, known drug-target biology was recovered (Fig. 4a): MEK inhibitors killed KRAS-mutant lines (AZD8330: $d = -0.62$, $q = 0.0007$; Trametinib: $d = -0.57$, $q = 0.0006$), mTOR and AKT inhibitors killed PTEN-mutant lines (Everolimus: $d = -0.61$, $q = 0.0009$; Ipatasertib: $d = -0.99$, $q = 0.0001$), and HDAC inhibitors killed EP300-mutant lines (Panobinostat in Ovarian cancer: $d = -1.45$, $q = 0.010$). These pharmacologic patterns add biological plausibility to the paralog-SL predictions but do not demonstrate paralog-specific targeting.

To distinguish context-selective paralogs from pan-essential ones, we defined a dependency window score (DWS; an in vitro dependency-selectivity ratio, not a clinical dose–toxicity therapeutic index):

$$\mathrm{DWS} = \frac{|\mathrm{DD}|}{\max(|\mu_{\mathrm{Chronos}}|,\, f_{\text{pan-essential}},\, 0.01)} \tag{2}$$

where $f_{\text{pan-essential}}$ is the fraction of cell lines with paralog Chronos score < -0.5. Each paralog was classified into one of four selectivity tiers (Fig. 4b). Two pairs fell into HIGH_SELECTIVITY: ARID1A→ARID1B (mean DWS $= 2.82$ across four cancer contexts; mean selectivity $+0.28$) and the established synthetic-lethal pair SMARCA4→SMARCA2 (4.87; $+0.18$), whose recovery acts as a positive control for the framework. This is what one would expect if ARID1A loss creates a specific dependency on ARID1B-containing BAF complexes [18]: within the dependency-window module ARID1A→ARID1B attained the highest mean selectivity and the second-highest mean $|\mathrm{DD}|$ (0.270) of all evaluated pairs (Supplementary Table S6). The highest raw DWS value was attained by NF1→RASA2 (5.42), but this estimate rests on a near-zero pan-essential denominator (0.1% of cell lines) with selectivity ≈ 0, so NF1→RASA2 is not a selective candidate. BRCA1↔BRCA2 and PIK3R1→CRKL were classified as PAN_ESSENTIAL (pan-essential fraction 0.53–0.55 and 0.73), suggesting that targeting them directly would cause broad cellular toxicity. Nine paralog pairs maintained DWS > 1.0 in ≥ 2 cancer contexts (Supplementary Fig. S7).

Structural analysis of 11 key paralog pairs showed that six (55%) had perfect domain architecture conservation (Fig. 4c). EP300↔CREBBP was the most structurally similar (0.976), consistent with shared histone acetyltransferase function and strong CPTAC co-variation. ARID1A→ARID1B ranked highest for PROTAC-based strategies (PROTAC score $= 0.670$), driven by high intrinsic disorder and lysine density — features that favor PROTAC-mediated degradation over conventional small-molecule binding. PIK3CA↔PIK3CB was the best small-molecule target (druggability $= 0.616$), reflecting its well-folded kinase domains. ESMFold-based pocket analysis of 15 proteins confirmed that compact, structured domains (KRAS/HRAS, FBXW2) yield the highest druggability scores, while large disordered proteins such as ARID1B are more suited to PROTAC approaches (Supplementary Fig. S10). A composite preclinical score integrating normalized DWS (0.40), normalized selectivity (0.30), and structure-based targetability (0.30) ranked NF1→RASA2 first (0.695) and ARID1A→ARID1B second (0.631; Fig. 4d; Table 2); the NF1 score is driven by the near-zero pan-essential denominator noted above (selectivity ≈ 0), so ARID1A→ARID1B remains the leading selective candidate.

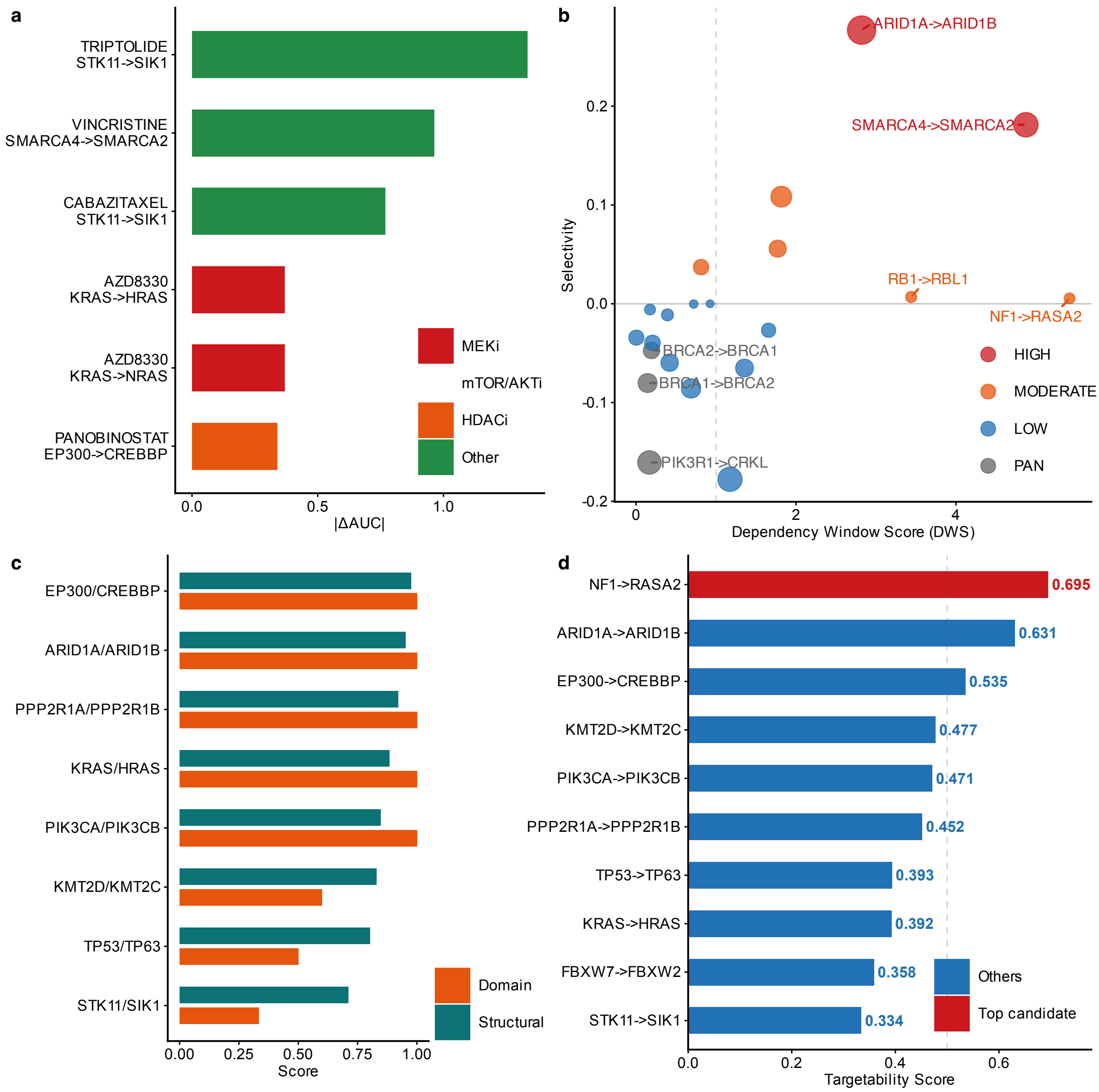

Fig. 4: Pharmacologic context and translational prioritization. **a**, PRISM drug selectivity: —ΔAUC— for top drug-paralog associations. Known drug-target biology is recapitulated: MEK inhibitors (AZD8330, Trametinib) selectively kill KRAS-mutant lines; mTOR/AKT inhibitors (Everolimus, Ipatasertib) kill PTEN-mutant lines; HDAC inhibitors (Panobinostat) kill EP300-mutant lines. BH $q < 0.25$. See Supplementary Fig. S5 for full results. **b**, Therapeutic window classification. Bubble size, mean DWS; color, selectivity tier. HIGH_SELECTIVITY requires selectivity > 0.15 and DWS > 1.0; NF1→RASA2 has high DWS (5.42) but near-zero selectivity (0.005), hence MODERATE. **c**, Structural similarity (top row) and domain conservation (bottom) for 11 key paralog pairs. Filled circles, perfect domain conservation (Jaccard $= 1.0$). EP300↔CREBBP has the highest structural similarity (0.976). **d**, Composite preclinical prioritization score ranking the top 10 candidates. Scores integrate normalized dependency window score (0.40), normalized selectivity (0.30), and structure-based targetability (0.30). NF1→RASA2 ranks first (0.695) through a near-zero pan-essential denominator that inflates its raw DWS (selectivity ≈ 0); ARID1A→ARID1B (0.631) is the leading selective candidate.

Table 2: Top paralog-SL candidates ranked by preclinical prioritization score. DWS: mean dependency window score across up to 5 cancer contexts on the ≥5-mutant frame. Class: HS = HIGH_SELECTIVITY, MOD = MODERATE, LS = LOW_SELECTIVITY, PE = PAN_ESSENTIAL. Struct: structural similarity score. Target.: composite preclinical score. Known SL pairs marked with ⋆. *NF1→RASA2's top composite score is driven by a near-zero pan-essential denominator (0.1% of cell lines) that inflates its DWS; with selectivity ≈ 0 it is not a selective candidate, and ARID1A→ARID1B remains the leading selective candidate. All candidates are computationally nominated and require experimental validation before clinical translation.*

Driver	Paralog	DWS	Class	Known	Struct	Target.
NF1	RASA2	5.42	MOD		0.658	0.695
ARID1A	ARID1B	2.82	HS	⋆	0.952	0.631
EP300	CREBBP	1.82	MOD	⋆	0.976	0.535
KMT2D	KMT2C	1.77	MOD		0.829	0.477
PIK3CA	PIK3CB	1.36	LS	⋆	0.847	0.471
PPP2R1A	PPP2R1B	0.93	LS	⋆	0.921	0.452
TP53	TP63	0.81	MOD		0.803	0.393
KRAS	HRAS	0.21	LS		0.885	0.392
FBXW7	FBXW2	0.72	LS	⋆	0.666	0.358
STK11	SIK1	0.39	LS	⋆	0.710	0.334

Discussion

We set out to test whether a single, interpretable metric — the dependency shift between driver-mutant and wild-type cells — could compete with black-box classifiers for paralog-SL prediction. Across 23 solid tumor types, the interpretable composite score reached AUROC = 0.831 in head-to-head comparison on 72 pairs, statistically indistinguishable from the best multi-feature classifier (SVM-RBF, 0.843; classifier range 0.114–0.843), while DD alone reached 0.566. The dependency shift carries a substantial but not dominant share of the signal; PCS and expression features contribute complementary information.

Why protein, not RNA? Our seven-cohort CPTAC survey found consistent paralog protein co-variation but a marginal RNA-level signal (AUROC = 0.564). EP300↔CREBBP was significant in five of seven cohorts; KRAS↔HRAS correlation reached $r = 0.667$ in LUSC. A plausible mechanism is complex stoichiometry: many paralog pairs (EP300/CREBBP, ARID1A/ARID1B) encode subunits of the same multi-subunit complex, and loss of one subunit can stabilize the remaining paralog by reducing competition for complex assembly or proteasomal degradation. This would produce protein-level but not transcript-level co-variation, exactly what we observed. The mutation-conditioned analysis in UCEC is consistent with this model: ARID1B protein was 55% higher in ARID1A-mutant tumors, though the result is suggestive ($p = 0.082$) and needs replication in additional cohorts and pairs. If the stoichiometric model holds, it also provides a rationale for PROTAC-based degradation strategies, which exploit protein abundance rather than transcriptional regulation.

TSG versus oncogene contexts. DD performed better in TSG-driven cancers (mean AUROC = 0.814) than oncogene-driven cancers (0.768) on the ≥3-per-group sensitivity frame. The difference did not reach significance (permutation $p = 0.645$), and with only three oncogene lineages the comparison is underpowered; the primary ≥5 frame retains a single oncogene-driven lineage, precluding estimation there. Biologically, however, the direction makes sense: TSG loss-of-function creates classical synthetic lethality where a paralog compensates for a missing function, whereas oncogene gain-of-function (KRAS, PIK3CA) may produce synthetic dosage lethality — a weaker, more context-dependent dependency shift. For practical purposes, DD is most reliable in TSG-driven contexts; oncogene-driven paralog dependencies likely require pathway-aware models.

Clinical implications. MSI status may modulate DD signal strength (MSI-H $\geq$ MSS in both colorectal and endometrial subgroups on the sensitivity frame; Fig. 3a), and truncating mutations elicited stronger compensation than missense variants for several TSG pairs. For an ARID1B-directed trial, these observations suggest a two-tier enrichment strategy — require ARID1A truncating mutations and select patients with low baseline ARID1B expression to maximize the therapeutic window — without restricting enrollment by MSI status. This is a computational proposal; the enrichment magnitude needs prospective estimation. Our DWS framework nominated ARID1A$\rightarrow$ARID1B (DWS = 2.82) as the leading selective candidate while flagging BRCA1/2 paralogs as pan-essential. The DWS is based entirely on in vitro data and cannot predict organ-specific in vivo toxicity; natural extensions would incorporate tissue expression data from GTEx, mouse knockout phenotypes from IMPC, and germline constraint metrics (pLI/LOEUF) from gnomAD.

Relationship to prior work. D'Antonio et al. [16] published the foundational analysis of paralog genetic interactions, and De Kegel et al. [43] systematically predicted robust paralog synthetic lethality across hundreds of cancer cell lines using interpretable features (protein interaction, evolutionary conservation). Combinatorial CRISPR platforms have since mapped paralog synthetic lethals experimentally [44, 46], converging on 13 candidate cross-study gold standards [44] whose replication across screens remains imperfect [45]. Our work is complementary to these efforts in three ways. We benchmark DD against eight published deep learning methods in the CV3 setting — a comparison their study did not perform. Our seven-cohort CPTAC proteomic survey reveals protein-level co-variation that their analysis did not examine. And we add clinically actionable stratification (MSI, mutation type) plus a quantitative therapeutic window framework for translational prioritization. Together, the five orthogonal evidence layers go beyond a gene-family dependency atlas toward a prioritized, experimentally actionable candidate list.

Data independence. Both DD and the gold-standard labels come from DepMap CRISPR data, which raises a circularity concern. We addressed this with a three-pronged strategy. First, the headline estimate does not depend on DepMap-derived or direction-unsupported labels: removing the two gold-standard pairs with DepMap-era evidence increased AUROC from 0.676 to 0.725, restricting to the eight pairs with pre-DepMap experimental evidence raised it to 0.774, and relabelling the pair whose direct evidence supports only the reciprocal direction (EP300$\rightarrow$CREBBP) as non-positive left it unchanged at 0.676, confirming the signal predates the dataset. Both lineage-evaluable positive entries on the Tier A $\cup$ Tier B external benchmark (ARID1A$\rightarrow$ARID1B in two lineages) ranked above all unlabeled controls; the remaining benchmark pairs had too few qualifying driver-mutant lines for evaluation. Second, CPTAC proteomics, PRISM drug sensitivity, and TCGA survival provide independent data layers. Third, prospective validation against SL pairs catalogued after the DepMap 26Q1 release is planned.

The ARID1A$\rightarrow$ARID1B paradox. Our top-ranked candidate has the largest lineage-level DD (0.386 in Ovarian cancer) and the leading dependency-window profile among novel selective candidates (DWS = 2.82; mean selectivity +0.28), yet its within-driver BH-corrected q-values remain nominally non-significant (0.39 in Ovarian, 0.88 in Endometrial). This reflects ARID1A's >200 HGNC paralogs creating an extreme multiple-testing burden, not weak signal. The q-value tests whether ARID1A→ARID1B is significantly enriched among all ARID1A paralogs; the multivariate regression ($p = 5.8 \times 10^{-42}$ after controlling for TP53 and CNV) tests whether ARID1B dependency shifts in ARID1A-mutant cells. For clinical prioritization, effect size and selectivity matter more than within-driver FDR ranking; the q-value highlights the gap between statistical significance and biological relevance when sample sizes are small.

Evaluation frameworks. We report two AUROC values. The cancer-type-specific evaluation (AUROC = 0.676; 110 driver×paralog×lineage entries, 8 positives) preserves lineage-specific DD values and is the primary framework. The per-pair aggregated evaluation (AUROC = 0.500; 72 unique pairs, 6 positives) averages |DD| across lineages per pair and, after the class-specific driver-mutation rules shrank the evaluable positive set, no longer exceeds the permutation null — a power limitation we report explicitly rather than a framework we rely on. Bootstrap and permutation analyses use the per-pair framework.

Limitations.

1. *Sample size:* Several rare driver mutations appear in fewer than five DepMap cell lines and could not be analyzed in the primary framework; some lineages lacked sufficient samples for MSI and mutation-type stratification.
2. *Gold-standard heterogeneity:* The evaluation set mixes functional analogs (e.g., BRCA1↔BRCA2, whose low k-mer Jaccard ≈ 0.23 reflects short 3-mers shared by chance between long proteins rather than true sequence homology) with true sequence paralogs. Stratifying by paralog type in future evaluations would improve clarity.
3. *Circular validation:* DD and the gold standard both derive from DepMap data; the core AUROC measures internal consistency, not independent prediction. CPTAC, PRISM, and TCGA provide partially independent support but do not directly validate SL prediction.
4. *Observational design:* DD captures association, not causation. We controlled for lineage, co-mutations, and CNV through multivariate regression but cannot eliminate residual confounding. CRISPR double-knockout in isogenic models remains the necessary next step.
5. *In vitro therapeutic window:* DWS is based on cell line dependencies and cannot predict organ-specific toxicity. Integration with GTEx, IMPC, and gnomAD constraint data is needed.
6. *Structural analysis:* Limited to sequence-level features. AlphaFold-Multimer complex predictions and experimental binding data are needed for structure-guided drug design.
7. *CPTAC coverage:* Not all paralogs were quantified in all cohorts (41–45 of 58 per cohort); mutation-conditioned analysis was performed only for ARID1A→ARID1B in UCEC.

Future directions. (1) CRISPR double-knockout validation of ARID1A→ARID1B in ARID1A-mutant versus wild-type isogenic lines (e.g., TOV-21G, HCT116, MCF7), with viability and colony formation as primary readouts. This study is purely computational; experimental validation through collaboration with functional genomics laboratories is the essential next step. (2) External hold-out validation against SL pairs catalogued in SynLethDB or other databases after the DepMap 26Q1 release. (3) Extending mutation-conditioned proteomic analysis from ARID1A→ARID1B in UCEC to additional paralog pairs and CPTAC cohorts. (4) PROTAC-based targeting of ARID1B, leveraging its high lysine density and intrinsic disorder. (5) Building lineage-specific paralog-SL atlases with MSI-aware filtering for clinical trial biomarker development.

Conclusions

We introduce Delta Dependency, a single-subtraction metric for paralog-based SL prediction, and evaluate it across 23 solid tumor types. DD achieves internal-consistency AUROC of 0.676 for the paralog-SL task within DepMap (1.000 on the Tier A $\cup$ Tier B external benchmark, on which two of five pairs were lineage-evaluable), and the interpretable composite score reaches 0.831; published deep learning results in an analogous CV3 setting remain higher (SLMGAE [14], 0.790) but are contextual reference points, not head-to-head benchmarks. We note that these evaluations use different test sets; a rigorous head-to-head benchmark would require running all methods on identical paralog-SL data. CPTAC proteomics across seven cohorts reveals protein-level paralog co-variation that is undetectable at the RNA level, and mutation-conditioned analysis in UCEC provides initial evidence for compensatory protein upregulation. MSI status and mutation type offer clinically actionable stratification biomarkers. Quantitative therapeutic window analysis combined with PRISM drug screening and structural druggability assessment nominates ARID1A→ARID1B as the leading computational candidate for preclinical development, pending experimental validation.

We provide a complete open-source toolkit — the `paralogSL` R package and a reproducible Python pipeline — that requires only publicly available DepMap data and no specialized computational infrastructure.

Methods

Data sources and preprocessing

DepMap 26Q1. CRISPRGeneEffect.csv (420 MB; Chronos-normalized gene dependency scores (hereafter "gene effect scores") for 1,208 cell lines $\times$ 18,531 genes), OmicsSomaticMutations.csv (554 MB), OmicsExpressionProteinCodingGenesTPMLogp1.csv (483 MB), and Model.csv (681 KB) were downloaded from the DepMap Portal (https://depmap.org/portal/download/, release 26Q1) [31, 32]. Driver-mutation status was defined by gene-class-specific rules on the default omics profile per model (`IsDefaultEntryForMod` [33]: for tumor suppressors (TSG), a line is mutant only if it carries a DepMap-curated likely loss-of-function variant (`LikelyLoF`, which includes dominant-negative TP53 missense); for oncogenes, only if it carries a recurrent oncogenic hotspot variant (`Hotspot`); genes were classified as TSG or oncogene by the DepMap 26Q1 `OncogeneHighImpact/TumorSuppressorHighImpact` majority vote (one documented override: GATA3 as TSG). Amplification-driven oncogenes without qualifying hotspot mutations (e.g., CCNE1) contribute no mutant lines under this rule and are analysed through copy-number channels instead. The complete per-gene variant classification table (old non-silent definition vs. class-specific rule, cell line counts, and variant-type distribution) is provided as Supplementary Table S7. Cell lines without expression or dependency data were excluded, yielding 1,208 cell lines. Cross-study dependency data integration followed established methods [34].

HGNC gene families. The HGNC complete set was downloaded from https://storage.googleapis.com/public-download-files/hgnc/tsv/tsv/. Protein-coding genes with "Approved" status and non-empty `gene_group_id` were retained (11,050 genes). All pairwise gene group combinations yielded 66,595 paralog pairs.

CPTAC proteomics. Ovarian HGSC global proteome data were obtained from the NCI Proteomic Data Commons (PDC study PDC000360, $n = 232$) [28]. Seven additional CPTAC cohorts were retrieved via the cBioPortal API [26, 27]: BRCA ($n = 122$), COAD ($n = 97$), LUAD ($n = 110$), GBM ($n = 99$), PDAC ($n = 140$), UCEC ($n = 95$), and LUSC ($n = 108$). Pearson correlations were computed on $\log_2$-transformed abundance values ($\geq$10 paired samples per test).

PRISM drug sensitivity. The PRISM Repurposing secondary screen dataset (1,482 compounds $\times$ 727 cell lines) was downloaded from the DepMap Portal [30]. $\Delta\text{AUC} = \text{mean}(\log_2 \text{AUC}_{\text{MUT}}) - \text{mean}(\log_2 \text{AUC}_{\text{WT}})$. Selective drugs: BH $q < 0.25$, ΔAUC< 0, enrichment> 0.1.

TCGA survival data. Pan-cancer overall survival and gene expression data were obtained from the UCSC Xena browser [29]. Breast cancer patients ($n = 1{,}082$) were stratified by median paralog expression. Cox proportional hazards models estimated hazard ratios.

Delta Dependency computation

DD was computed as defined in Equation 1, using Welch's t-test (unequal variances); effect sizes are reported as Cohen's d (pooled SD) and the small-sample-corrected Hedges' g. The primary analysis required a minimum of five mutant and five wild-type cell lines per driver$\times$lineage stratum; the more permissive $\geq$3/$\geq$3 threshold was re-run as a sensitivity analysis (Results). Benjamini-Hochberg correction [35] ($q < 0.20$) was applied within each driver gene: for driver D, all (D, P_i, c_j) combinations were corrected jointly. This is a discovery-stage pipeline; global FDR correction across all 66,595 pairs $\times$ 40 drivers $\times$ 23 cancer types was not performed because the goal was to maximize sensitivity for candidate identification. Candidates should be treated as hypothesis-generating. At a stricter $q < 0.10$, PIK3CA$\rightarrow$PIK3CB remained significant ($q = 0.086$ in Endometrial), whereas ARID1A$\rightarrow$ARID1B did not survive within-driver correction ($q = 0.50$), not because of weak signal ($|\text{DD}| = 0.30$) but because ARID1A has >200 HGNC paralogs.

Benchmark comparison

Gold-standard positives and matched negatives. Twelve paralog-SL pairs were curated from published studies (Supplementary Table S3), which assigns each pair an evidence tier after verification of the primary citations. *Tier A* (three pairs: AKT1$\rightarrow$AKT2 [38], CDK4$\rightarrow$CDK6 and MAP2K1$\rightarrow$MAP2K2 [46]) comprises pairs with direct genetic synthetic-lethal evidence from dual-gene perturbation (combinatorial or digenic CRISPR knockout). *Tier B* (two pairs: SMARCA4$\rightarrow$SMARCA2 [17], ARID1A$\rightarrow$ARID1B [18]) comprises pairs for which the paralog is a demonstrated selective dependency in driver-mutant cells (natural-genotype conditional dependency from single-gene perturbation with functional validation). The Tier A $\cup$ Tier B set (five pairs) constitutes the primary external benchmark. *Tier C* (five pairs) comprises indirect evidence only and is excluded from the primary benchmark: EP300$\rightarrow$CREBBP, established experimentally in the reciprocal direction only (CREBBP$\rightarrow$EP300 [20, 41]); PIK3CA$\rightarrow$PIK3CB, whose primary evidence supports PTEN$\rightarrow$PIK3CB [22]; CCNE1$\rightarrow$CCNE2, supported by developmental redundancy in mouse double knockouts [39]; and FBXW7$\rightarrow$FBXW2 and PPP2R1A$\rightarrow$PPP2R1B, derived from DepMap analyses. Two mechanistic comparators (BRCA1$\leftrightarrow$BRCA2 [4] and STK11$\rightarrow$SIK1, an LKB1–SIK axis reference [40]) are not sequence paralogs and serve as specificity references only. Of the twelve pairs, 10 are true sequence paralogs (shared gene family, conserved domains), 1 is a partial homolog (STK11→SIK1), and 1 is a functional analog (BRCA1↔BRCA2). Unlabeled controls comprised all remaining HGNC-defined paralog pairs involving the same 40 driver genes that did not appear in the positive set; these controls are unlabeled rather than experimentally confirmed non-SL pairs, so the evaluation is a positive–unlabeled setting and AUROC quantifies discrimination against unlabeled controls, not against verified negatives. The primary ranking score is $|\mathrm{DD}|$ (Equation 1 defines the signed quantity); because $|\mathrm{DD}|$ also promotes reverse-direction effects (dependency decreasing after driver mutation, e.g., oncogene-addiction displacement for PIK3CA→PIK3CB), we audited every evaluation frame for directionality (`output/direction_audit.json`): on the Tier A ∪ Tier B benchmark set, signed and absolute scoring coincide (AUROC = 1.000), whereas on the full curated set 3 of 8 positive entries have DD < 0 (signed-DD AUROC = 0.629 vs. $|\mathrm{DD}|$ 0.676). Full-set estimates should therefore be read as direction-agnostic discrimination, and directional claims rest on the Tier A ∪ Tier B benchmark. The evaluation is highly imbalanced (12 curated positives vs. ∼194 unlabeled controls per lineage). AUROC is insensitive to class prevalence, but precision–recall performance is not; we therefore report AUPRC (average precision) alongside AUROC, with bootstrap confidence intervals for both. Directional pairs (e.g., BRCA1→BRCA2 and BRCA2→BRCA1) are scored separately when both driver genes are in the 40-gene panel.

Contextual comparison with published methods.

DD was compared against eight published SL prediction methods evaluated in the CV3 setting by Feng et al. [13]. Published AUROCs are from the CV3 evaluation reported in Supplementary Data 1 of that study (NSMRand negative sampling, 1:1 positive:negative ratio, complete dataset); DD was evaluated on the paralog-SL prediction task using the 12 gold-standard paralog-SL pairs as the positive set. AUROC was computed with the rank-based Mann–Whitney statistic (identical to scikit-learn 1.3.0 [36]); AUPRC is reported as average precision. Bootstrap 95% CIs used 10,000 resamples of evaluation entries (or pairs, for per-pair frames) with replacement; paired bootstrap resamples were used for head-to-head comparison of $|\mathrm{DD}|$ against component scores on identical resamples. For the DD + identity-proxy subset (3 pairs), bootstrap CI was computed separately to account for the reduced sample size. Negative controls: 100 permutations of shuffled known-SL labels.

Sequence identity computation

K-mer-based Jaccard similarity ($k = 3$) was used as a computationally efficient proxy for global protein sequence identity. This metric was validated against Needleman-Wunsch global alignment identity in a subset of 50 protein pairs (13 known paralog pairs and 37 non-paralog controls; Pearson $r = 0.88$, $p < 10^{-16}$; Supplementary Fig. S9). Within-group correlations were $r = 0.91$ (paralogs, $n = 13$) and $r = 0.41$ (non-paralogs, $n = 37$), confirming that the high overall r is not solely driven by between-group variance. The $\geq$30% threshold should therefore be interpreted as an approximate filter for enriching high-identity pairs rather than a precise sequence identity cutoff.

Potential confounders and causal interpretation

DD quantifies an observational association between driver mutation status and paralog dependency, not a causal effect. Potential confounders include: (1) lineage differences between mutant and wild-type cell lines within a cancer type; (2) co-occurring mutations that may independently affect paralog dependency; (3) copy number alterations at paralog loci. We mitigated these through: within-lineage analysis (DD computed separately per cancer type), Welch's t-test (robust to unequal variances), and CNV independence verification (Supplementary Fig. S4, all $R^2 < 0.10$). Benjamini-Hochberg correction was applied within each driver gene to control false discovery rate. Nevertheless, experimental validation (e.g., CRISPR double-knockout in isogenic cell lines) remains essential to establish causality.

CPTAC proteomic co-variation

Protein abundance data for seven CPTAC cohorts (BRCA, COAD, LUAD, GBM, PDAC, UCEC, LUSC; $n = 771$ total) were retrieved via the cBioPortal API [26, 27]. For each paralog pair, Pearson correlation of protein $\log_2$-abundance was computed across all samples within each cohort. Significance was assessed using BH correction within each pair across 7 cohorts. Mutation-conditioned proteomic analysis was performed for the ARID1A→ARID1B pair in the UCEC cohort: ARID1A mutation status was retrieved from the cBioPortal mutations API (`ucec_cptac_2020_mutations`), and ARID1B protein abundance was compared between ARID1A-mutant ($n = 37$) and wild-type ($n = 58$) samples using the Wilcoxon rank-sum test. Extension to additional pairs and cohorts is a priority future direction.

Pan-cancer lineage definitions

Twenty-seven solid tumor types were defined by matching OncotreePrimaryDisease annotations to curated keyword lists; the 23 types with ≥ 6 cell lines with available data were analyzed (Supplementary Table S1). Lineage short labels follow DepMap/Oncotree conventions and mix organ-based and histology-based names as required by each cancer type's classification (e.g., NSCLC and SCLC are distinct histological subtypes of lung cancer); full names are given in Supplementary Table S1. Forty pan-cancer driver genes were selected from TCGA PanCanAtlas driver analyses [23] and the COSMIC Cancer Gene Census [42], filtered to genes with ≥ 3 mutant cell lines in at least one lineage (Supplementary Table S5).

MSI stratification

Cell lines were classified with the official DepMap 26Q1 microsatellite annotation (`OmicsGlobalSignatures.csv`). The DepMap consortium computes an MSIsensor2 MSIscore from each model's sequencing data; following the consortium convention, lines with MSIscore > 20 were labelled MSI-H and all others MSS. Default-entry profiles (`IsDefaultEntryForModel`) were used; all 87 evaluable colorectal and endometrial lines had official annotations. Because the annotation is measured rather than mutation-inferred, POLE-mutant lines receive their observed status (15 of 22 POLE-mutant lines in these two lineages are officially MSI-H) and are not presumed hypermutated; agreement between this annotation and a damaging-mutation proxy (MLH1/MSH2/MSH6/PMS2/POLE) was 89% (per-line audit at `output/msi_classification_audit.csv`). DD was computed separately within each subgroup for colorectal and endometrial cancers.

Mutation type classification

Driver mutations classified as truncating (frameshift, nonsense, splice-acceptor, splice-donor, start-loss, stop-loss) or missense (missense_variant, protein_altering_variant, excluding truncating) using VEP VariantInfo annotations. Per-cell-line classification used the most severe mutation type.

Therapeutic window analysis

DWS defined as in Equation 2. The denominator uses $\max(|\mu_{\text{Chronos}}|, f_{\text{pan-essential}}, 0.01)$ to capture the most conservative estimate of baseline paralog essentiality: $|\mu_{\text{Chronos}}|$ reflects mean dependency magnitude (dimensionless, normalized by Chronos), $f_{\text{pan-essential}}$ reflects the fraction of lines where the paralog is essential (dimensionless, range 0–1), and the 0.01 floor prevents division by near-zero values for non-essential paralogs. All three quantities are dimensionless and on comparable scales (0–1); the max operator selects whichever indicates the greatest baseline toxicity risk. Paralogs classified: PAN_ESSENTIAL ($f_{\text{pan-essential}} > 0.5$), HIGH_SELECTIVITY (selectivity > 0.15 and DWS > 1.0), MODERATE, or LOW_SELECTIVITY. Analysis across 5 cancer contexts (Ovarian, Endometrial, Breast, Colorectal, PanCancer).

Structural druggability

Sequence-based features (length, MW, pI, GRAVY hydrophobicity, disorder propensity, lysine/cysteine density) from UniProt canonical sequences. Structural similarity: $1 - \overline{|\Delta\text{feature}|}$. Domain architectures from UniProt annotations. PROTAC suitability: lysine_density $\times$ 2.0 $+$ disorder_propensity $-$ cysteine_density $\times$ 0.5. Composite targetability: dependency window score (0.40), selectivity (0.30), structural similarity (0.15), druggability (0.15).

Statistical analyses

AUROC via scikit-learn 1.3.0 [36]. Bootstrap CI: 2.5th and 97.5th percentiles from 1,000 iterations. Negative controls: 10,000 label permutations. Analyses in Python 3.10 with numpy 1.24 [37], scipy 1.11 [47], pandas 2.0, and statsmodels 0.14. The companion R package `paralogSL` provides DD computation, PCS analysis, benchmark comparison, and CPTAC correlation visualization under the MIT license.

Power analysis and test set stratification

Gold-standard pairs were stratified into evidence tiers (Supplementary Table S3): *Tier A* (3 pairs; direct dual-perturbation genetic synthetic-lethal evidence), *Tier B* (2 pairs; natural-genotype conditional dependency with functional validation), *Tier C* (5 pairs; indirect, reciprocal-direction-only, or DepMap-derived evidence), and 2 mechanistic comparators (BRCA1$\leftrightarrow$BRCA2, STK11$\rightarrow$SIK1). The pre-specified primary endpoint is the AUROC and AUPRC of |DD| on the Tier A $\cup$ Tier B external benchmark at the driver$\times$paralog$\times$lineage level — the natural unit at which DD is computed — with the pair treated as the grouping unit in sensitivity analyses to avoid cross-lineage information leakage. The full 12-pair curated set, tier-restricted, direction-strict, and per-pair aggregated analyses are reported as secondary and sensitivity analyses. With at most 2–5 evaluable positive pairs, subset-level AUROC has limited power:

at $\alpha = 0.05$, an estimated $\geq$25 validated positive pairs would be needed to achieve 80% power for detecting AUROC = 0.85 against a null of 0.5. To compensate for this limitation, we complement binary classification metrics with effect-size ranking (—DD—, DWS, selectivity) and multivariate regression significance (p-values for individual pairs).

Supplementary information

Supplementary Figures S1–S10 and Supplementary Tables S1–S8 are available as a separate PDF file. Supplementary Fig. S1: Panoramic baseline across 23 solid tumor types (cell line counts and tested pairs, colored by AUROC tier). Supplementary Fig. S2: Cross-cancer DD AUROC detailed values. Supplementary Fig. S3: CPTAC per-cohort scatter plots. Supplementary Fig. S4: CNV independence analysis. Supplementary Fig. S5: PRISM drug selectivity full results. Supplementary Fig. S6: Full mutation-type stratification. Supplementary Fig. S7: Therapeutic window full classification. Supplementary Fig. S8: Full bootstrap distribution and negative control. Supplementary Fig. S9: k-mer Jaccard vs. Needleman-Wunsch alignment identity validation ($n = 50$ pairs, Pearson r shown). Supplementary Fig. S10: Structure-based druggability assessment of 15 proteins using ESMFold-predicted structures. Supplementary Table S1: Cell line counts per cancer type. Supplementary Table S2: Complete paralog-SL pair analysis results (110 driver$\times$paralog$\times$cancer-type associations passing the minimum sample size filter of $\geq$5 mutant and $\geq$5 wild-type cell lines). Supplementary Table S3: Gold-standard paralog-SL pair evidence. Supplementary Table S4: Pan-cancer summary statistics. Supplementary Table S5: Complete 40-gene driver panel with cancer types, selection source, and functional category. Supplementary Table S6: All 21 paralog pairs ranked by dependency window score with component values. Supplementary Table S7: Gene-class-specific driver-mutation rules and per-gene variant classification (TSG: LikelyLoF; oncogene: Hotspot). Supplementary Table S8: Per-association effect sizes (Cohen's d, Hedges' g) and Welch t-test p-values for all 110 evaluated associations (TSV file).

Abbreviations

SL: synthetic lethality; DD: Delta Dependency; PCS: paralog co-dependency score; DWS: dependency window score; TSG: tumor suppressor gene; LoF: loss of function; CPTAC: Clinical Proteomic Tumor Analysis Consortium; TCGA: The Cancer Genome Atlas; PRISM: Profiling Relative Inhibition Simultaneously in Mixtures; DepMap: Cancer Dependency Map; AUROC: area under the receiver operating characteristic curve; AUPRC: area under the precision–recall curve; MSI-H: microsatellite instability-high; MSS: microsatellite stable; MMR: mismatch repair; CNV: copy number variation; HNSCC: head and neck squamous cell carcinoma; SCLC: small cell lung cancer; NSCLC: non-small cell lung cancer; OR: odds ratio; HR: hazard ratio; CI: confidence interval; LR: logistic regression; RF: random forest; SVM: support vector machine.

Declarations

Ethics approval and consent to participate

Not applicable. This study used exclusively publicly available, de-identified data.

Consent for publication

Not applicable.

Data availability

All data are publicly available: DepMap 26Q1 from https://depmap.org/portal/download/; CPTAC proteomics from https://cbioportal.org and https://pdc.cancer.gov; TCGA PanCan from https://xenabrowser.net; PRISM from https://depmap.org/portal/prism/. HGNC gene families from https://www.genenames.org/. Processed datasets with cell line-level DD values and all supplementary tables are archived at Zenodo (DOI: to be assigned upon acceptance) and the GitHub repository below, enabling direct integration with clinical genomic pipelines.

Code availability

All analysis code is available under the MIT license at https://github.com/tjogzt/paralogSL (DOI: 10.5281/zenodo.21502113). The complete analysis pipeline (DOI: 10.5281/zenodo.21502030) is available at https://github.com/tjogzt/paralog-sl-predictor. The `paralogSL` R package (v1.1.0, 19 exported functions) provides DD computation, benchmark comparison, therapeutic window analysis, CPTAC visualization, and clinical decision support (`predict_trial_response`, `visualize_dependency_shi`
Random seeds (`set.seed(42)` in R, `random_state=42` in Python) are fixed throughout.
Workflow: DepMap gene-effect matrix → `build_mutation_matrix()` → `compute_dd()` per driver×paralog×lineage → `compute_auroc()` against gold standard → `compute_therapeutic_window()` (historical function name retained in the package; computes the DWS) for translational ranking → visualization via `plot_pancancer_summary()` and `plot_therapeutic_window()`.
Quick start (3 lines):

```
install.packages("paralogSL")  # or devtools::install_github("taozhu/paralogSL")
library(paralogSL)
result <- run_paralog_analysis(dep_matrix, mut_matrix, driver="ARID1A", paralog="ARID1B")
```

The package includes built-in datasets (`known_sl_pairs`, `gyn_drivers`, `solid_tumor_summary`), a vignette with worked examples, and 19 exported functions. A frozen release with all processed data tables is archived at Zenodo (https://doi.org/10.5281/zenodo.21502030). The complete Python analysis pipeline (`main.py`, `pcs.py`, `pancancer.py`) is included in the same repository for full reproducibility. In addition, four audit scripts (`compute_headline_metrics.py`, `ml_benchmark.py`, `regression_controls.py`, and `audit_manuscript_numbers.py`) recompute every quantitative claim in this manuscript directly from raw DepMap data and derived artifacts — headline metrics, classifier benchmarks, covariate-adjusted regressions, and all remaining numbers (lineage-level AUROCs, mechanism contrasts, MSI and mutation-type stratifications, direction audit, therapeutic-window module, and machine-readable table mirrors), 121 claims in total — each ending with an automated claim-by-claim comparison against the reported values; a single entry point (`verify_all.sh`) runs all four audits plus the test suite in under one minute.

Competing interests

The authors declare no competing interests.

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Authors' contributions

Q.Q.M. and T.Z. conceived the study. T.Z. developed the methodology, performed all computational analyses, and developed the R package. Q.Q.M. curated clinical datasets and provided gynecological oncology domain expertise. Q.Q.M. and T.Z. wrote the manuscript.

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